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CS 621 DATA ANALYTICS

Introduction

WE FLY

UNDER DIFFERENT NAMES

- Data Analytics
- Data Science
- Big Data
- Data-Intensive Computing
- ...



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WHY THIS COURSE?

- Lots of data – Big Data !!
- Many new applications and challenges.
- CS/Software focused.

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SOME BIG NUMBERS

- 90% of the world's data created in the last two years
- 500 million tweets a day
- 100 hours of video uploaded to YouTube every minute
- 426 items sold per second by Amazon (Cyber Monday)
- 2.5 billion pieces of content are shared on Facebook every day
- >100 billion emails, 55 million photos by instagram, ...
- These numbers are getting even larger as we speak.

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NEW CHALLENGES

- Lots of Data – varieties in formats, all sorts of contents, huge quantity
- Data accumulated so fast – streaming data
- Immediate insight – real time or near r.t.
- Scalability/Availability
 - Data is distributed, computation is also distributed, ...
- Need many talents, e.g., Programmers, Data Scientists, and Context Experts
- ...

CHARACTERISTICS OF BIG DATA

- **3Vs:**
 - **Volume** – lots of Data (TB, PB, ..)
 - **Velocity** – accumulated fast
 - **Variety** – lots of information/knowledge
- 3Vs + Value and Veracity

DRIVEN BY

- Hardware (Storage, Devices, Sensors, Networking)
- Software, Internet of Things
- Applications in various domains
 - Business
 - Natural Science
 - Social Science
 - Health Care
 - Education
 - ...

WHO WORKS WITH BIG DATA?

- People in specific domains, e.g. Banking.
- Statisticians
- Applied Math people
- CS people
- Business people
- IT people
- ...



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